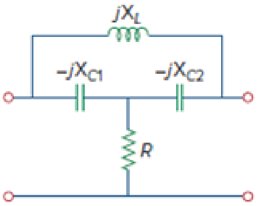
Chapter 19, Problem 44P

Problem

Using Fig. 19.99, design a problem to help other students better understand how to find the transmission parameters of an ac circuit.

Figure 19.99



Step-by-step solution

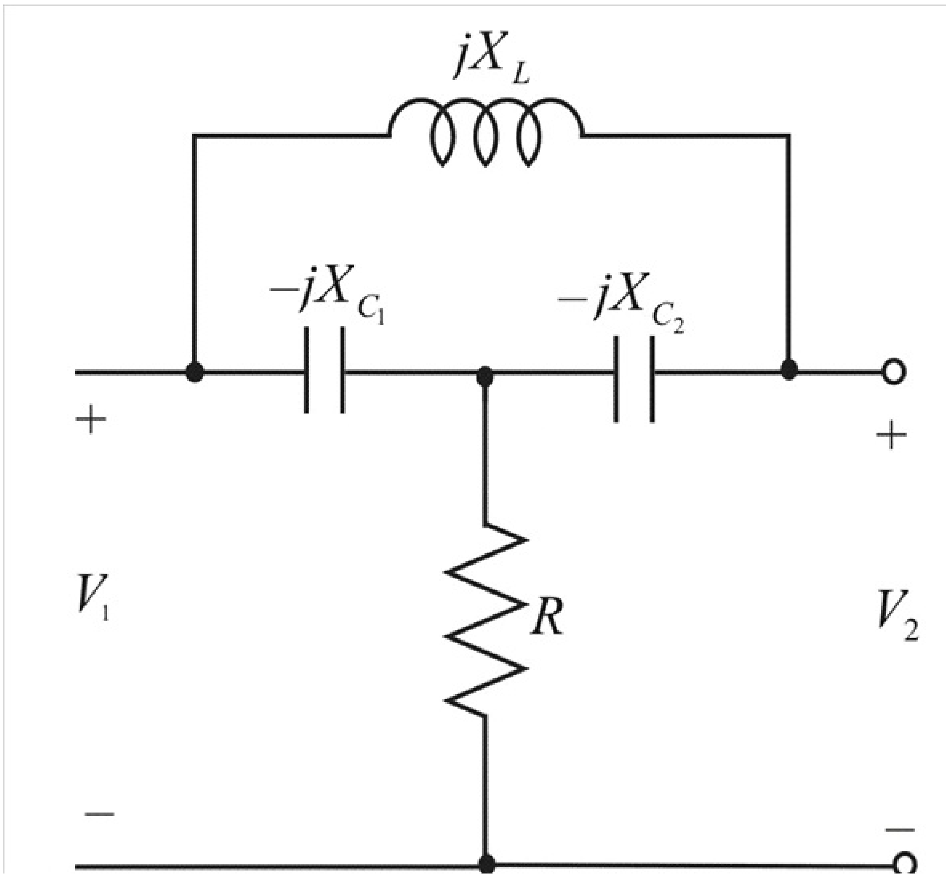
Step 1 of 12

The transmission parameters

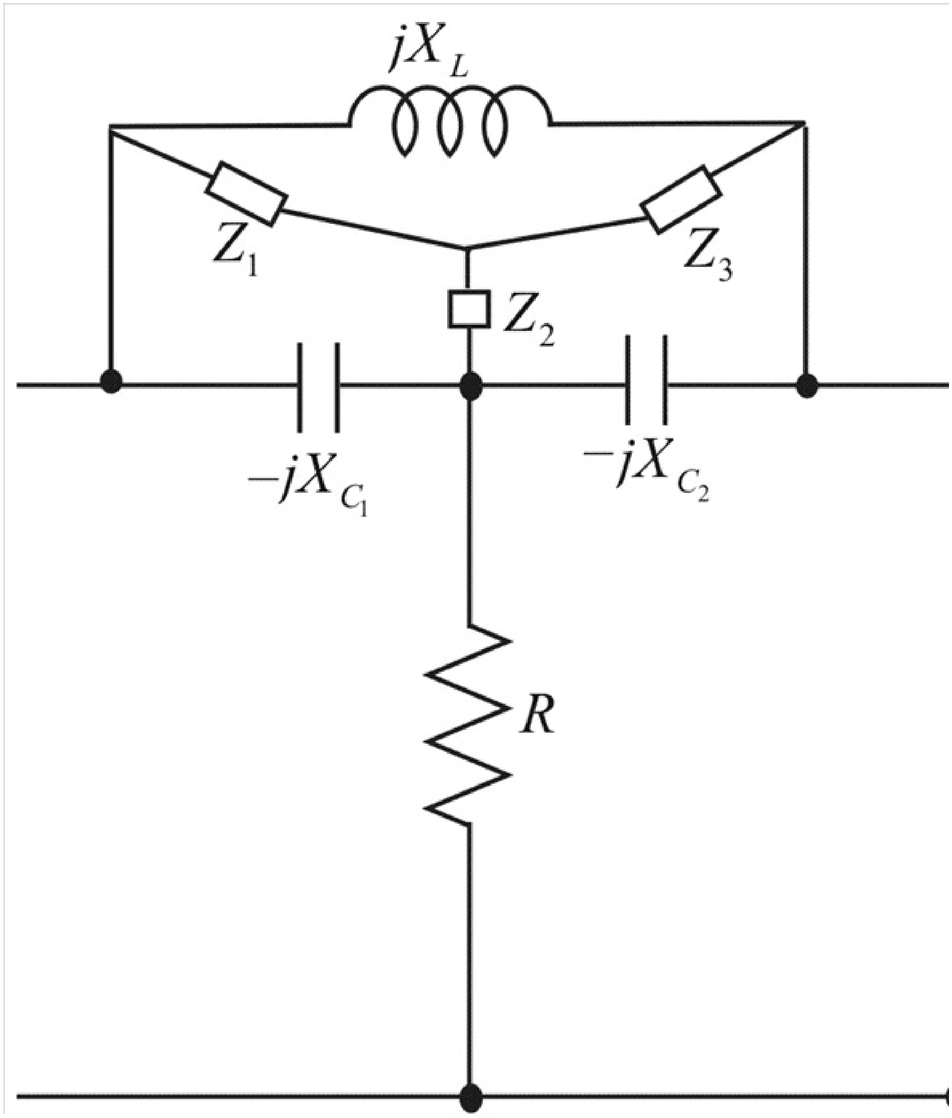
$$V_1 = AV_2 - BI_2$$

$$I_1 = CV_2 - DI_2$$

Step 2 of 12



Step 3 of 12



Step 4 of 12

Convert Δ to T network, then

$$Z_1 = \frac{(jX_L)(-jX_{C_2})}{jX_L - jX_{C_1} - jX_{C_2}}$$

$$Z_1 = \frac{X_L X_{C_1}}{j\left(X_L - X_{C_1} - X_{C_2}\right)}$$

$$Z_2 = \frac{\left(-jX_{C_1}\right)\left(-jX_{C_2}\right)}{j\left(X_L - X_{C_1} - X_{C_2}\right)}$$

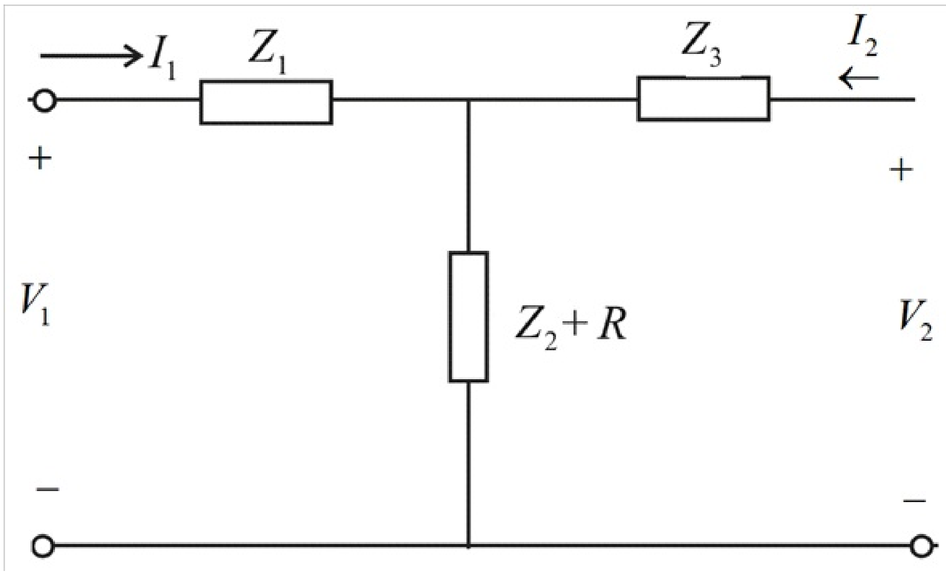
$$Z_2 = \frac{-X_{C_1} X_{C_2}}{j\left(X_L - X_{C_1} - X_{C_2}\right)}$$

$$Z_3 = \frac{(jX_L)\left(-jX_{C_2}\right)}{j\left(X_L - X_{C_1} - X_{C_2}\right)}$$

$$Z_3 = \frac{X_L X_{C_2}}{j\left(X_L - X_{C_1} - X_{C_2}\right)}$$

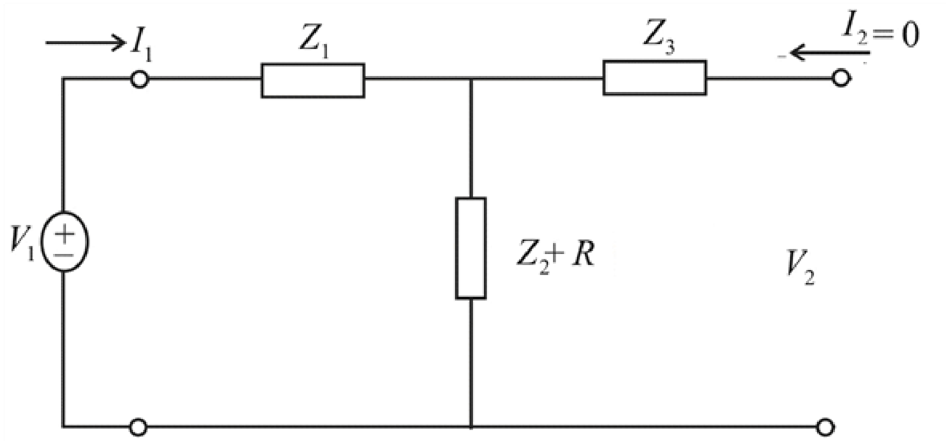
Step 5 of 12

Thus, the circuit becomes



Step 6 of 12

To determine A and C, we open the output port, so that $I_2 = 0$ and place a voltage source V_1 at the input port



By voltage division rule

$$V_2 = \frac{Z_2 + R}{Z_1 + Z_2 + R} V_1$$

$$\frac{V_1}{V_2} = \frac{Z_1 + Z_2 + R}{Z_2 + R}$$

$$\frac{V_1}{V_2} = 1 + \frac{Z_1}{Z_2 + R}$$

Step 7 of 12

$$V_2 = 1 + \frac{\dfrac{X_L X_{C_1}}{j\left(X_L - X_{C_1} - X_{C_2}\right)}}{\dfrac{-X_{C_1} X_{C_2}}{j\left(X_L - X_{C_1} - X_{C_2}\right)} + R}$$

$$V_2 = 1 + \frac{X_L X_{C_1}}{-X_{C_1} X_{C_2} + jR\left(X_L - X_{C_1} - X_{C_2}\right)}$$

We know that

$$A = \left.\frac{V_1}{V_2}\right|_{I_2=0}$$

$$\therefore A = 1 - \frac{X_L X_{C_1}}{X_{C_1} X_{C_2} + jR\left(X_{C_1} + X_{C_2} - X_L\right)}$$

Step 8 of 12

$$V_2 = \left(Z_2 + R\right) I_1$$

$$\frac{I_1}{V_2} = \frac{1}{\left(Z_2 + R\right)}$$

$$\frac{I_1}{V_2} = \frac{1}{\dfrac{-X_{C_1} X_{C_2}}{j\left(X_L - X_{C_1} - X_{C_2}\right)} + R}$$

$$\frac{I_1}{V_2} = \frac{j\left(X_L - X_{C_1} - X_{C_2}\right)}{-X_{C_1} X_{C_2} + jR\left(X_L - X_{C_1} - X_{C_2}\right)}$$

$$\frac{I_1}{V_2} = \frac{\left(X_L - X_{C_1} - X_{C_2}\right)}{jX_{C_1} X_{C_2} + R\left(X_L - X_{C_1} - X_{C_2}\right)}$$

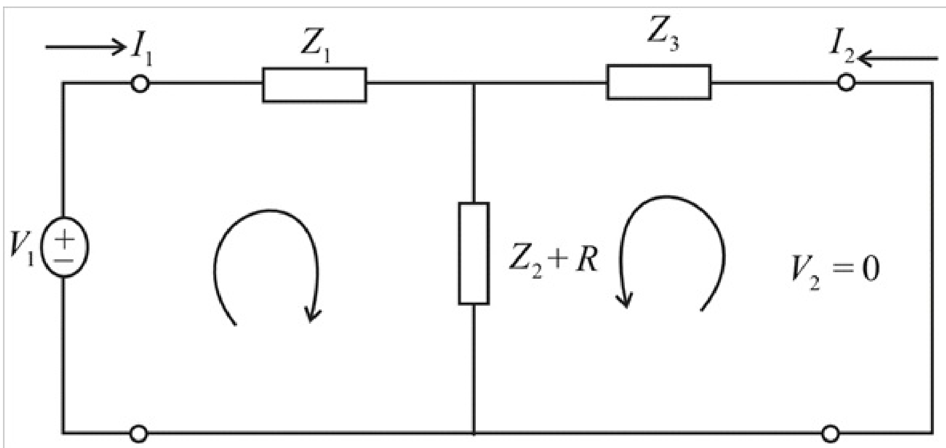
We know that

$$C = \left.\frac{I_1}{V_2}\right|_{I_2=0}$$

$$\therefore C = \frac{X_{C_1} + X_{C_2} - X_L}{-jX_{C_2} X_{C_1} + R\left(X_{C_1} + X_{C_2} - X_L\right)}$$

Step 9 of 12

To obtain B and $D,$ we short circuit the output port so that $V_2 = 0$ and place a voltage source V_1 at the input port.



By current division rule

$$I_2 = -\frac{Z_2 + R}{Z_2 + Z_3 + R} I_1$$

$$I_2 = -\frac{\dfrac{-X_{C_1}X_{C_2}}{j\left(X_L - X_{C_1} - X_{C_2}\right)} + R}{\dfrac{-X_{C_1}X_{C_2}}{j\left(X_L - X_{C_1} - X_{C_2}\right)} + \dfrac{X_L X_{C_2}}{j\left(X_L - X_{C_1} - X_{C_2}\right)} + R} I_1$$

$$I_2 = -\frac{-X_{C_1}X_{C_2} + jR\left(X_L - X_{C_1} - X_{C_2}\right)}{-X_{C_1}X_{C_2} + X_L X_{C_2} + jR\left(X_L - X_{C_1} - X_{C_2}\right)} I_1$$

$$-\frac{I_1}{I_2} = \frac{-X_{C_1}X_{C_2} + X_L X_{C_2} + jR\left(X_L - X_{C_1} - X_{C_2}\right)}{-X_{C_1}X_{C_2} + jR\left(X_L - X_{C_1} - X_{C_2}\right)}$$

We know that

$$D = \frac{-I_1}{I_2}$$

$$\therefore D = \frac{-X_{C_1}X_{C_2} + X_L X_{C_2} + jR\left(X_L - X_{C_1} - X_{C_2}\right)}{-X_{C_1}X_{C_2} + jR\left(X_L - X_{C_1} - X_{C_2}\right)}$$

To find the equivalent input resistance

$$R_{eq} = Z_1 + \left(Z_2 + R\right) \parallel Z_3$$

$$R_{eq} = Z_1 + \frac{\left(Z_2 + R\right)Z_3}{\left(Z_2 + R\right) + Z_3}$$

$$R_{eq} = \frac{V_1}{I_1}$$

$$\frac{V_1}{I_1} = Z_1 + \frac{\left(Z_2 + R\right)Z_3}{\left(Z_2 + R\right) + Z_3}$$

$$V_1 = \frac{Z_1\left(\left(Z_2 + R\right) + Z_3\right) + \left(Z_2 + R\right)Z_3}{\left(Z_2 + R\right) + Z_3} I_1$$

$$I_1 = -\left[\frac{Z_2 + Z_3 + R}{Z_3 + R}\right] I_2$$

$$\frac{I_1}{I_2} = -\left[\frac{Z_2 + Z_3 + R}{Z_3 + R}\right]$$

$$V_1 = \frac{Z_1\left(\left(Z_2 + R\right) + Z_3\right) + \left(Z_2 + R\right)Z_3}{\left(Z_2 + R\right) + Z_3} I_1$$

$$V_1 = -\frac{Z_1\left(\left(Z_2 + R\right) + Z_3\right) + \left(Z_2 + R\right)Z_3}{\left(Z_2 + R\right) + Z_3}\left[\frac{Z_2 + Z_3 + R}{Z_3 + R}\right] I_2$$

$$-\frac{V_1}{I_2} = \frac{Z_1\left(\left(Z_2 + R\right) + Z_3\right) + \left(Z_2 + R\right)Z_3}{Z_3 + R}$$

$$B=\frac{\frac{X_L X_{C_1}}{j\left(X_L-X_{C_1}-X_{C_2}\right)}\left(\left(\frac{-X_{C_1}X_{C_2}}{j\left(X_L-X_{C_1}-X_{C_2}\right)}+R\right)+\frac{X_L X_{C_2}}{j\left(X_L-X_{C_1}-X_{C_2}\right)}\right)+\left(\frac{-X_{C_1}X_{C_2}}{j\left(X_L-X_{C_1}-X_{C_2}\right)}+R\right)\frac{X_L X_{C_2}}{j\left(X_L-X_{C_1}-X_{C_2}\right)}}{\frac{X_L X_{C_2}}{j\left(X_L-X_{C_1}-X_{C_2}\right)}+R}$$

$$B=\frac{X_L X_{C_1}\left(\left(-X_{C_1}X_{C_2}+jR\left(X_L-X_{C_1}-X_{C_2}\right)\right)+X_L X_{C_2}\right)+\left(-X_{C_1}X_{C_2}+jR\left(X_L-X_{C_1}-X_{C_2}\right)\right)X_L X_{C_2}}{j\left(X_L X_{C_2}+jR\left(X_L-X_{C_1}-X_{C_2}\right)\right)\left(X_L-X_{C_1}-X_{C_2}\right)}$$